



Time to reclassify? The prognostic value of radiologic splenic vessel involvement in resectable pancreatic ductal adenocarcinoma of the body and tail: a systematic review and meta-analysis

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Background: The prognostic impact of radiographic splenic vessel involvement in pancreatic ductal adenocarcinoma (PDAC) of the body and tail remains controversial.

Methods: A systematic search was performed using PubMed, Embase, the Cochrane Library, Web of Science, and Scopus in accordance with the PRISMA and AMSTAR guidelines. The primary outcome was 5-year overall survival (OS). Time-to-event method was used.

Results: Eight articles involving 1678 patients were analyzed. Patients with radiological splenic artery (SA) invasion had poorer OS than those without [hazard ratio (HR) 1.61, 95% confidence interval (CI) 1.14–2.29; $P = 0.007$]. Similarly, radiological splenic vein (SV) involvement was associated with significantly worse OS (HR 1.66, 95% CI 1.13–2.44; $P = 0.010$). Subgroup analyses for SA involvement, categorized by “abutment” and “encasement,” showed an overall HR of 2.35 (95% CI 1.72–3.19; $P < 0.00001$), with HRs of 1.77 (95% CI 1.25–2.59; $P = 0.001$) for abutment and 3.03 (95% CI 2.20–4.17; $P < 0.00001$) for encasement, indicating a significant difference between subgroups ($P = 0.02$). For SV involvement, the overall HR was 2.15 (95% CI 1.49–3.10; $P < 0.001$), with HRs of 1.66 (95% CI 0.97–2.84; $P = 0.07$) for abutment and 2.80 (95% CI 1.98–3.96; $P < 0.0001$) for encasement, showing no significant difference between subgroups ($P = 0.11$).

Conclusion: This meta-analysis showed worse survival in patients with splenic vessel infiltration of PDAC. Therefore, SA and SV invasion may serve as indications for neoadjuvant therapy for resectable PDAC.

Keywords: pancreatic ductal adenocarcinoma, prognosis, splenic artery, splenic vein, neoadjuvant therapy

Introduction

By 2040, pancreatic ductal adenocarcinoma (PDAC) is projected to become the second leading cause of cancer-related deaths in the US, surpassing colorectal cancer with an estimated 46 000 annual

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HIGHLIGHTS

- This meta-analysis evaluates the prognostic impact of radiologic splenic artery (SA) and vein involvement in resectable body and tail PDAC.
- Both SA and vein invasion are independently associated with worse overall survival.
- SA encasement is a strong negative prognostic factor in resectable pancreatic cancer.
- These findings challenge the current resectability criteria based on vessel involvement.
- The results support considering neoadjuvant therapy in patients with isolated splenic vessel invasion.

fatalities^[1]. Surgical resection is the only curative treatment; nevertheless, only 20–25% of patients are candidates for surgery^[2,3]. According to the National Comprehensive Cancer Network (NCCN) clinical practice guidelines, PDAC can be classified as resectable, borderline resectable (BR) and locally advanced (LA), based on the relationship between the tumor and its surrounding major vessels, such as the superior mesenteric artery/vein, portal vein (PV), coeliac axis and common hepatic artery on preoperative radiographic imaging^[4]. Tumors that abut or encase these vessels are associated with technical unresectability, high R1 rates, and early recurrence^[5,6]. Neoadjuvant therapy (NAT) has become the standard treatment for BR/LA PDAC to increase the rate of surgical resection and improve the postoperative prognosis^[7,8].

Notably, PDAC located in the body/tail is more likely to invade splenic vessels, which is associated with worse outcomes. However, radiological splenic vessel involvement is not considered a high-risk feature of aggressive biology or an indication for NAT in preoperative resectability evaluation, as tumors with splenic vessel invasion are technically resectable^[4].

Although pathological splenic vessel involvement has been previously reported to be a significant prognostic factor for PDAC located in the body/tail, postoperative histopathological confirmation cannot inform preoperative therapeutic decisions^[9–12]. This critical limitation has driven recent investigations into preoperatively assessable radiological indicators, with studies examining imaging-defined splenic vessel involvement, though critical uncertainties remain unresolved^[13–19].

For instance, contemporary studies have demonstrated marked discrepancies in vessel-specific prognostic significance^[13–20]. While several studies have examined splenic artery (SA) and vein involvement, their conclusions remain inconsistent, likely due to methodological heterogeneity and limited sample sizes. This uncertainty underscores the need for a systematic review and meta-analysis to provide more definitive evidence.

Therefore, by synthesizing data from multiple studies with larger sample sizes, this systematic review and meta-analysis aimed to comprehensively evaluate whether radiographic splenic vessel involvement in resectable pancreatic body/tail cancer holds oncological significance as a preoperative prognostic factor for upfront surgery, while systematically comparing the differential prognostic impact between SA and SV involvement. To ensure compliance with artificial intelligence reporting standards, this study adheres to the TITAN Guidelines 2025^[21].

Methods

Search strategy

We systematically searched PubMed, Embase, the Cochrane Library, Web of Science, and Scopus from inception to August 2025. The predefined search strategy combined Medical Subject Headings (MeSH) and free-text terms related to *pancreatic ductal adenocarcinoma* and *splenic vessels*. The search was limited to English-language publications. The detailed search strategy is presented in Supplemental Digital Content Material 1, available at: <http://links.lww.com/JS9/F858>. The reference lists of the selected articles were further examined to identify the relevant articles during the initial search. Only comparative clinical studies with full-text descriptions were included in this meta-analysis. The final inclusion of articles was determined by consensus of two independent researchers. Discrepancies when screening the title, abstract, and full text were resolved through discussion with senior researchers. This systematic review was conducted in accordance with the PRISMA and AMSTAR guidelines^[22,23]. Our protocol was registered in the International Prospective Register of Systematic Reviews (registration number: CRD420250655019).

Inclusion and exclusion criteria

Studies were required to meet all the following criteria: (1) Adults (≥ 18 years old) with histologically confirmed PDAC were localized to the body/tail of the pancreas. (2) Radiologically or surgically staged as resectable, according to the NCCN Guidelines^[4]. (3) Curative-intent distal pancreatectomy. (4)

Preoperative imaging (contrast-enhanced CT) evaluating SA/SV involvement. (5) Reported at least one of the following with extractable data: overall survival (OS), disease-free survival (DFS), recurrence-free survival (RFS), and progression-free survival (PFS). (6) Prospective/retrospective cohort studies with ≥ 50 patients and ≥ 12 months of follow-up.

Studies were excluded if any of the following applied. (1) Insufficient data for survival analysis despite contacting the authors. (2) High risk of bias (ROB) ($>30\%$ loss to follow-up without adequate handling). (3) Non-peer-reviewed reports without full methodology. (4) For duplicate reports from the same cohort, only the most comprehensive dataset was retained. (5) Patients who received NAT without separate HR data for those undergoing upfront surgery were included.

Outcomes

The primary outcome was 5-years' OS. As the similar definition of DFS, RFS, and PFS, and limited data of studies included on each of them, we integrate them into the final secondary outcome as "DFS/RFS/PFS."

Data extraction

Data from each eligible article were independently extracted in duplicates by two researchers. The extracted data included the first author, publication year, region of the study object, number of participants, mean age, and outcomes.

The Newcastle–Ottawa Scale (NOS) was used to assess the quality of each study in three dimensions: selection of the study population, comparability of the groups under study, and assessment of outcome. Studies that received a score of 9 stars were considered to have a low ROB, 7 or 8 stars to have a moderate ROB, and ≤ 6 stars to have a high ROB^[24]. All the studies included gained scores over 7.5 (Supplemental Digital Content Table S1, available at: <http://links.lww.com/JS9/F858>).

Statistical analysis

The meta-analysis was conducted following the Cochrane Handbook guidelines using Review Manager software (RevMan) version 5.4 for sensitivity analyses and funnel plot asymmetry tests. *P* values was two-sided, and $P < 0.05$ was considered significant. From each study, the HR and 95% confidence interval (CI) were retrieved for the association between SA/SV invasion and OS and DFS/RFS/PFS. When HR was not presented in the full text, it was estimated based on data available in the manuscript (number of patients exposed, patients at risk, and corresponding survival at different time points) using an Excel spreadsheet developed by Tierney *et al*^[25]. Heterogeneity was assessed using random effects models, with I^2 greater than 50% considered statistically significant. If I^2 was greater than 50%, a random-effects model was used. Publication biases were assessed by visual inspection of the funnel plots. Additionally, publication bias was assessed through visual inspection of funnel plots^[26]. A sensitivity analysis was performed by deleting each study individually.

Results

Identification and selection of reviews

The systematic literature search initially identified 368 records from five databases: PubMed ($n = 65$), Embase ($n = 89$),

Cochrane Library ($n = 8$), Web of Science ($n = 90$), and Scopus ($n = 116$) until August 2025. After removing 187 duplicates, 181 unique records underwent title and abstract screening, of which 165 were excluded because they were irrelevant to the research question. In addition, two records were identified through citation tracking. Therefore, a total of 18 full-text articles were assessed for eligibility. Following a detailed evaluation, a total of 10 studies were excluded from the analysis. Specifically, four studies were removed due to the absence of survival hazard ratio (HR) data stratified by vascular involvement, while another four were excluded because survival outcomes were reported based on pathological rather than radiological findings. In addition, one study was excluded owing to a small sample size, and one study was excluded because patients had received NAT prior to imaging. Ultimately, eight studies met all the inclusion criteria and were included in the quantitative meta-analysis (Fig. 1).

Characteristic of the included reviews

The included studies were conducted across various regions, including Japan, the USA, South Korea, and China, with sample sizes ranging from 68 to 454. Most of the studies were single-center retrospective cohorts, except for one multicenter retrospective cohort study by Yin *et al* in 2021^[18].

Outcome metrics mainly focused on OS, with some studies reporting DFS/RFS/PFS. The classification of vessel involvement varied across studies, with some using “clear/abutment vs. encasement” and others using “G1(clear), G2(abutment), G3(encasement)” (Fig. 2).

The HR for OS was reported in each study, with some studies showing statistically significant results. The median follow-up

period varied across studies from 17.3 to 57 months, with some studies not reporting this specific information^[18,19] (Table 1).

Findings of the included studies

We extracted the main results of the included studies focusing on the impact of splenic vessel involvement in PDAC of the body and tail. HR and CI for OS and DFS/RFS/PFS were provided for both SA and SV involvement in the multivariate and univariate analyses (Tables 2 and 3).

Long-term survival outcomes

Meta-analysis of long-term survival (6 studies) showed that patients with radiological SA invasion had worse 5-year OS than those without SA invasion (HR 1.61, 95% CI 1.14–2.29; $P = 0.007$; $I^2 = 69\%$) (Fig. 3A). There was high heterogeneity ($I^2 = 69\%$) in this analysis. Hence, sensitivity analysis was performed by deleting each study individually. (Supplemental Digital Content Figure S1, available at: <http://links.lww.com/JS9/F858>). Notably, the heterogeneity decreased to 25% after excluding the study by Mizumoto, which is partially due to its unique vessel invasion classification and the small number of SA involvement-positive patients. The funnel plot for publication bias and Egger's test ($P = 0.994$) indicated no evidence of publication bias (Supplemental Digital Content Figure S2, available at: <http://links.lww.com/JS9/F858>).

Similarly, patients with radiologically confirmed SV infiltration had a significantly poorer 5-year OS than those without SV infiltration (HR 1.66, 95% CI 1.13–2.44; $P = 0.010$) (Fig. 3B). The heterogeneity ($I^2 = 57$) for this analysis was high, and a sensitivity analysis was performed as previously

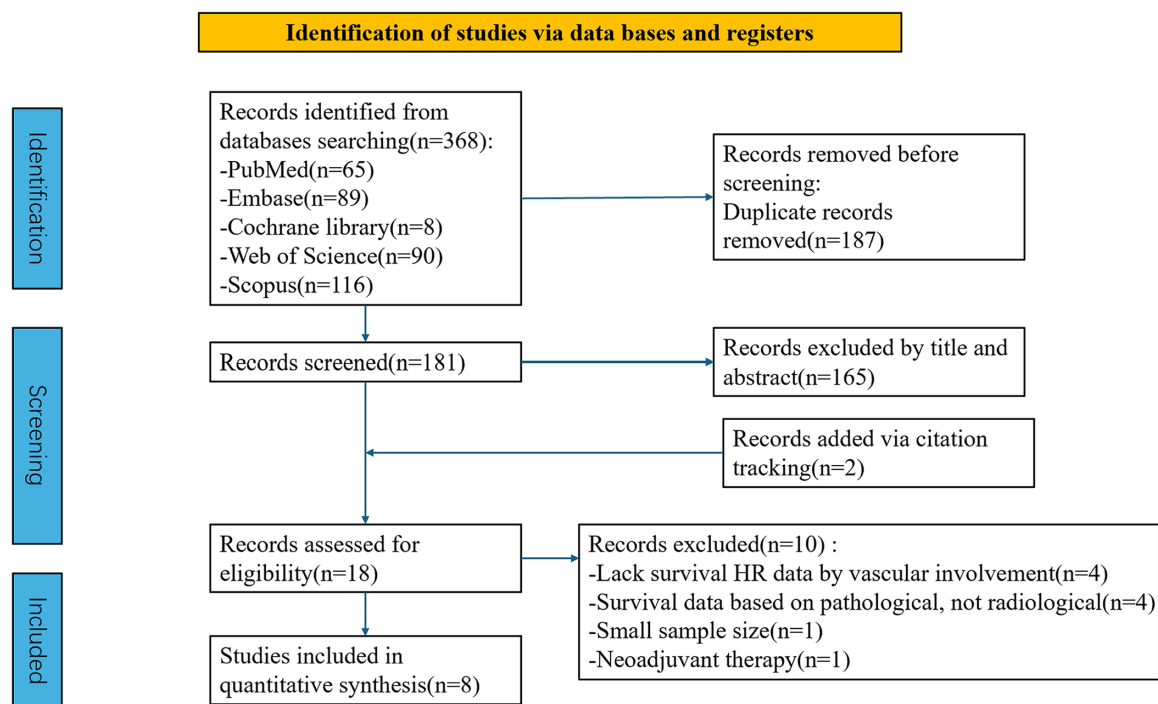


Figure 1. PRISMA flow diagram of study inclusion in the review.

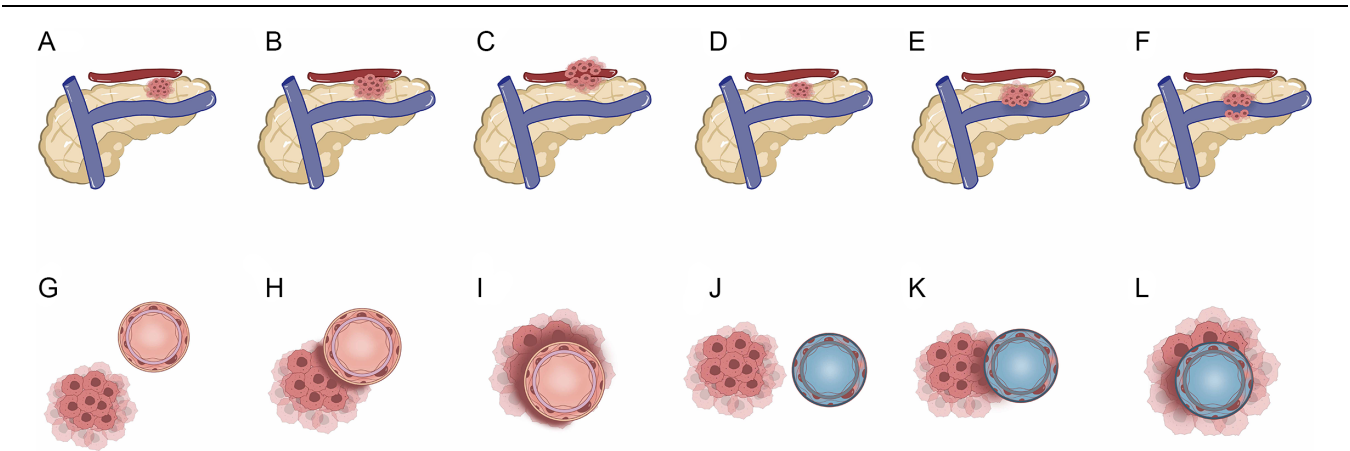


Figure 2. Schematic representation of splenic artery and vein invasion in PDAC. (A–F) Anterior anatomical views; (G–L) corresponding invasion angle diagrams. (A, G) Splenic artery (SA) clear; (B, H) SA abutment ($\leq 180^\circ$); (C, I) SA encasement ($>180^\circ$); (D, J) Splenic vein (SV) clear; (E, K) SV abutment($\leq 180^\circ$); (F, L) SV encasement ($>180^\circ$).

described (Supplemental Digital Content Figure S3, available at: <http://links.lww.com/JS9/F858>). By deleting each study individually, high heterogeneity persisted (52–64%) due to classification differences, but the funnel plot and Egger’s test ($P = 0.671$) showed low publication bias risk (Supplemental Digital Content Figure S4, available at: <http://links.lww.com/JS9/F858>).

Secondary outcomes

Analysis of the secondary outcomes showed that patients with radiological SA invasion had significantly worse DFS/RFS/PFS than those without SA invasion (HR = 2.10, 95% CI: 1.37–3.22, $P = 0.0007$; Fig. 4A). Moderate heterogeneity was observed ($I^2 = 53\%$). In contrast, no significant difference in DFS/RFS/PFS was found between patients with and without radiological SV infiltration (HR = 0.96, 95% CI: 0.39–2.34, $P = 0.92$), although the analysis exhibited considerable heterogeneity ($I^2 = 80\%$) (Fig. 4B).

Subgroup analysis of vascular invasion patterns

As the degree of vasculature serves as a major factor contributing to prognostic discrepancies, we further analyzed discrepancies between two vascular invasion patterns: “abutment” ($\leq 180^\circ$ contact) and “encasement” ($> 180^\circ$ contact) in the univariate analysis.

The findings of SA involvement revealed that encasement of the SA was associated with a higher HR (3.03, 95% CI 2.20–4.17; $P < 0.00001$) compared to abutment (HR 1.77, 95% CI

1.25–2.50; $P = 0.001$). The difference between encasement and abutment was statistically significant ($P = 0.02$; $I^2 = 80.1\%$), indicating a notably higher risk of encasement (Fig. 5A).

Similarly, in the analysis of SV involvement, encasement demonstrated a higher HR (2.80, 95% CI 1.98–3.96; $P < 0.00001$) than abutment (HR 1.66, 95% CI 0.97–2.84; $P = 0.07$), although the difference between these subgroups was not statistically significant ($P = 0.11$; $I^2 = 61.3\%$) (Fig. 5B).

Methodological quality of the included reviews

We assessed all the studies by quality assessment using NOS, with scores ranging from 7.5 to 9, reflecting high article quality (Supplemental Digital Content Table S1, available at: <http://links.lww.com/JS9/F858>).

For instance, certain studies had smaller sample sizes or specific patient selection criteria that may not fully represent the broader population of patients with PDAC in the body/tail^[13,16,17]. Additionally, follow-up quality was a notable area of improvement in several studies. While most studies reported adequate follow-up durations, some experienced higher rates of loss to follow-up or did not provide detailed reasons for loss, which could introduce bias and affect the reliability of the results^[14–19].

Discussion

This systematic review and meta-analysis evaluated the prognostic significance of splenic vessel involvement in resectable PDAC

Table 1 Characteristics of included systematic review					
Author	Year	Region	Study design	Sample size	Median follow-up (months)
Mizumoto <i>et al</i>	2018	Japan	Single-center retrospective cohort	68	27.8
Hyun <i>et al</i>	2019	USA	Single-center retrospective cohort	71	17.3
Yin <i>et al</i>	2021	USA	Multicenter retrospective cohort	454	/
Kawai <i>et al</i>	2021	Japan	Single-center retrospective cohort	102	21.4
Kang <i>et al</i>	2022	South Korea	Single-center retrospective cohort	249	27.8
Kitamura <i>et al</i>	2022	Japan	Single-center retrospective cohort	242	30
Xie <i>et al</i>	2023	China	Single-center retrospective cohort	258	/
Tan <i>et al</i>	2023	China	Single-center retrospective cohort	234	57

Table 2
Main results of included studies in multivariate analysis

Author	Year	OS			
		SA		SV	
		HR	95% CI	HR	95% CI
Mizumoto <i>et al</i>	2018	1.030	0.850–4.620	2.800	1.320–6.210
Hyun <i>et al</i>	2019	2.246	1.118–4.513		
Yin <i>et al</i>	2021			1.120	0.700–1.800
Kawai <i>et al</i>	2021	4.640	1.770–12.100	3.180	1.330–7.640
Kang <i>et al</i>	2022	1.520	1.020–2.030	1.130	0.720–1.530
Kitamura <i>et al</i>	2022	1.660	0.930–2.380		
Xie <i>et al</i>	2023				
Tan <i>et al</i>	2023	1.480	0.890–2.080	1.840	1.150–2.520

patients. Both SA and SV involvement are independent risk factors for OS. However, SA had a greater effect on DFS/PFS/RFS than SV did. Subgroup analysis revealed that encasement had a more detrimental prognostic impact than abutment. First, recent studies on whether SA or SV involvement is associated with a more severe prognosis remain controversial. For example, Kang *et al* identified SA involvement as an independent predictor of poor OS, while SV invasion showed no significance in their single-center cohort^[16]. Conversely, Mizumoto *et al* reported that SV infiltration, but not SA involvement, correlated with increased mortality risk^[13]. Some studies suggest that patients with SA involvement have a worse prognosis because the invasion of the SA makes it easier for cancer to disseminate^[11]. This may also be attributed to anatomical factors; tumors invading the SA tend to be located farther posteriorly, making complete surgical clearance more challenging and increasing the likelihood of positive retroperitoneal margins^[9,11]. Moreover, the dense perivascular neural plexus surrounding the SA may facilitate metastatic spread via perineural invasion, a route independent of vascular dissemination that has been associated with poor survival outcomes in pancreatic cancer^[11,27,28]. On the other hand, some studies hold the view that the involvement of the SV leads to a poorer prognosis, as it increases the likelihood of liver metastasis^[13,29]. A growing body of evidence supports the hypothesis that splenic or PV invasion facilitates hematogenous spread of tumor cells, particularly to the liver, via circulating tumor cells (CTCs). CTCs detected in the PV system have been shown to correlate with

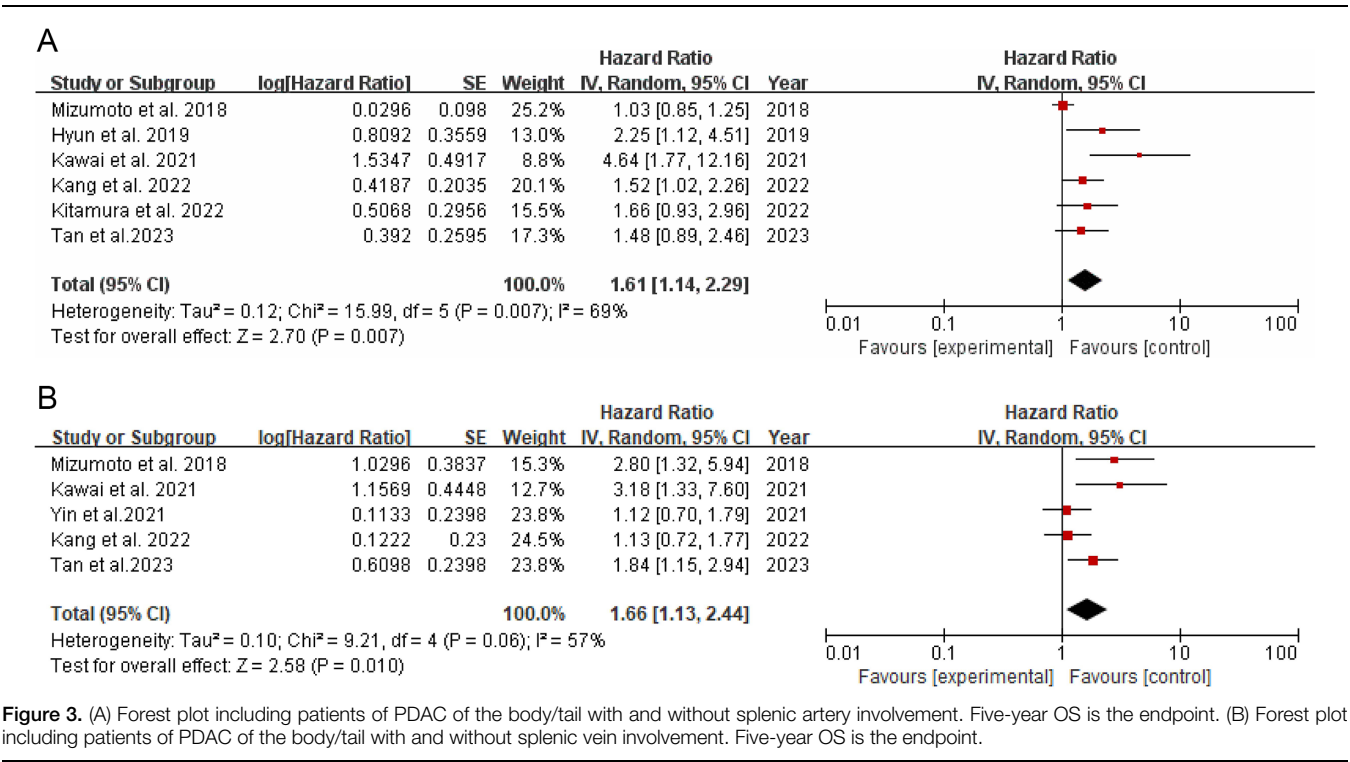
early liver metastasis and poorer outcomes in PDAC of the body and tail^[29,30]. Since the SV is the primary drainage route for these regions of the pancreas into the portal system, its invasion may allow tumor cells to directly access the systemic circulation. Tumors invading major vessels, such as the SV or PV, may thus shed more CTCs, increasing the risk of micrometastases and recurrence following surgery^[31,32]. These findings further highlight the need for more effective systemic therapies and support the consideration of NAT in select cases with vascular involvement. Recent studies reinforce this perspective. For example, Gantois *et al* demonstrated in a multicenter cohort that splenic vessel involvement and tumor size ≥ 30 mm were strongly associated with poor survival after distal pancreatectomy^[33]. Moreover, updated systematic reviews and meta-analyses have shown that NAT improves OS and R0 resection rates compared with upfront surgery in resectable pancreatic cancer^[34,35]. These contemporary findings strengthen our argument that radiographically resectable tumors with splenic vessel invasion may warrant a more aggressive neoadjuvant approach.

However, the comparable prognoses observed in this study may reflect opposing conclusions across the included studies, ultimately balancing out the meta-analysis. Second, consistent with previous studies, we found that patients with SA invasion had significantly worse DFS/RFS/PFS, whereas the SV group showed no significant difference^[14,18,20]. Third, consistent with Hyun's previous study, our subgroup analysis revealed that the difference between SA encasement and abutment was significant, yet no such difference was observed for SV^[17]. Although in most of the literature the HR for encasement is higher than that for abutment, the difference in SV encasement and abutment was not significant^[14,15].

Crucially, our study extends these findings by providing subtype-specific evidence for SA/vein invasion, a subgroup that is not explicitly stratified according to prior guidelines^[4]. While the NCCN framework broadly recommends NAT for vascular invasion, our data clarify that even “resectable” tumors with splenic vessel involvement (without mesenteric vessel involvement) exhibit a poor prognosis as BR subtypes. This supports a more nuanced application of NAT indications, in which splenic vascular invasion alone may warrant preoperative therapy to mitigate the high risk of micro-residual disease. The guidelines' emphasis on NAT for high-risk resectable tumors has been further validated in several studies. For example, Lof *et al* demonstrated in a multicenter retrospective cohort of distal

Table 3
Main results of included studies in univariate analysis

Author	Year	OS				DFS/PFS/RFS			
		SA		SV		SA		SV	
		HR	95% CI	HR	95% CI	HR	95% CI	HR	95% CI
Mizumoto <i>et al</i>	2018	2.160	1.060–4.110	3.490	1.860–6.830				
Hyun <i>et al</i>	2019	1.820	1.030–3.216	1.393	0.801–2.422				
Yin <i>et al</i>	2021	1.250	0.910–1.710	1.550	1.160–2.080	1.470	1.040–2.070	1.450	1.050–2.010
Kawai <i>et al</i>	2021	4.340	2.210–8.560	2.900	1.470–5.720	2.650	1.200–5.850	2.590	1.210–5.560
Kang <i>et al</i>	2022	1.640	1.150–2.340	1.620	1.110–2.380				
Kitamura <i>et al</i>	2022	2.320	0.950–3.690	1.680	0.530–2.830	2.670	1.350–4.000	2.330	1.210–3.460
Xie <i>et al</i>	2023							2.600	1.620–3.580
Tan <i>et al</i>	2023	2.340	1.220–3.470	2.500	1.570–3.440	2.450	0.680–4.220	2.480	0.500–4.470

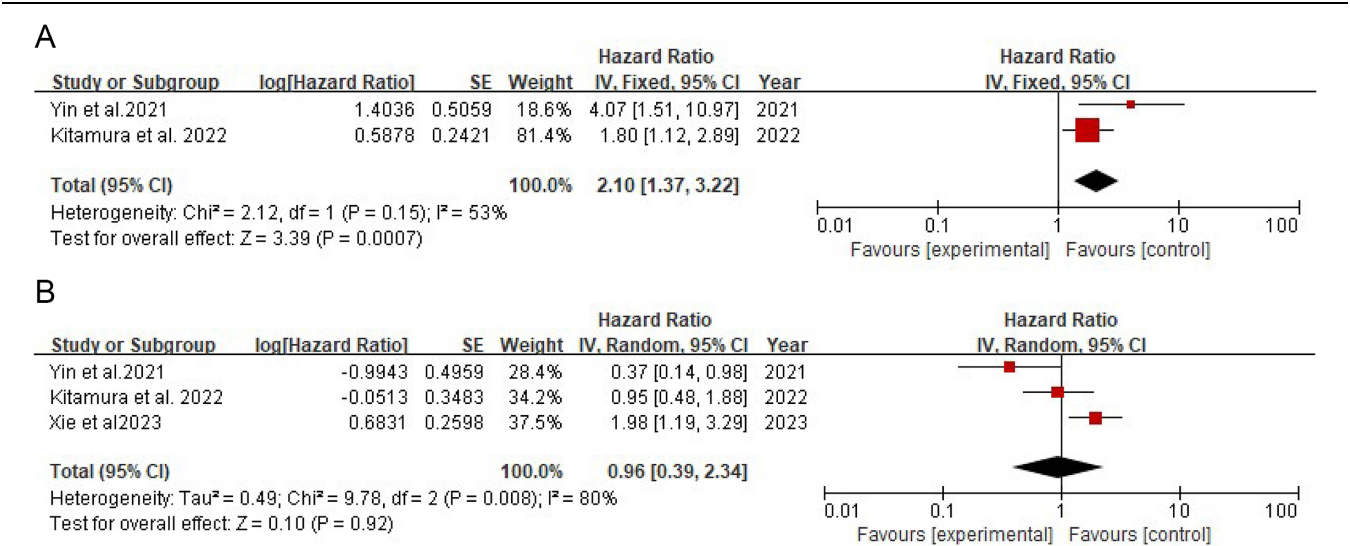


pancreatectomy for PDAC patients that NAT ($n = 37$, 14.7%) was associated with prolonged OS [36 months (95% CI 18–53) vs. 20 months (95% CI 15–24), $P = 0.049$], as compared with upfront resection in the sensitivity analysis of 251 patients with radiographic tumor involvement of splenic vessels^[36].

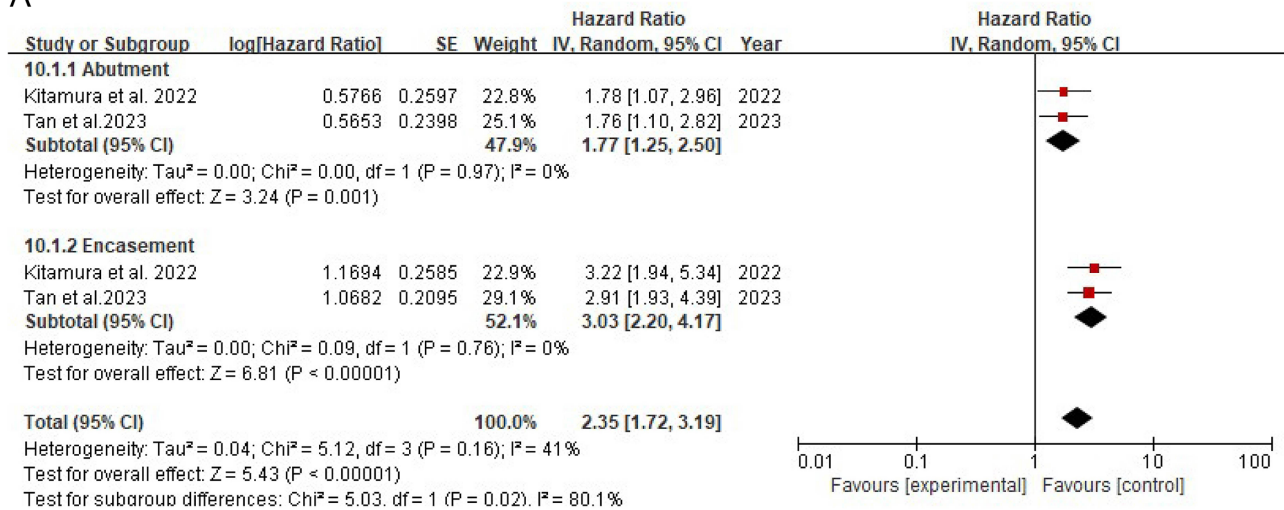
Notably, in three of the included studies, SA and SV involvement were stronger predictors of OS than tumor size or T stage, further supporting the argument that radiographically resectable tumors with splenic vessel invasion may harbor biologically

aggressive behavior akin to BR disease, suggesting the necessity of NAT in patients with splenic vessel involvement^[13,15,20].

This study had several notable strengths. First, compared with previous research, such as Crippa’s study, radiological assessment in our research offers more clinical predictive value^[37]. While Crippa’s study might rely on the histological evaluation of splenic vessels obtained from specimens, which is limited in preoperative application, our radiological approach can provide timely information before surgery. Second, the inclusion criteria



A



B

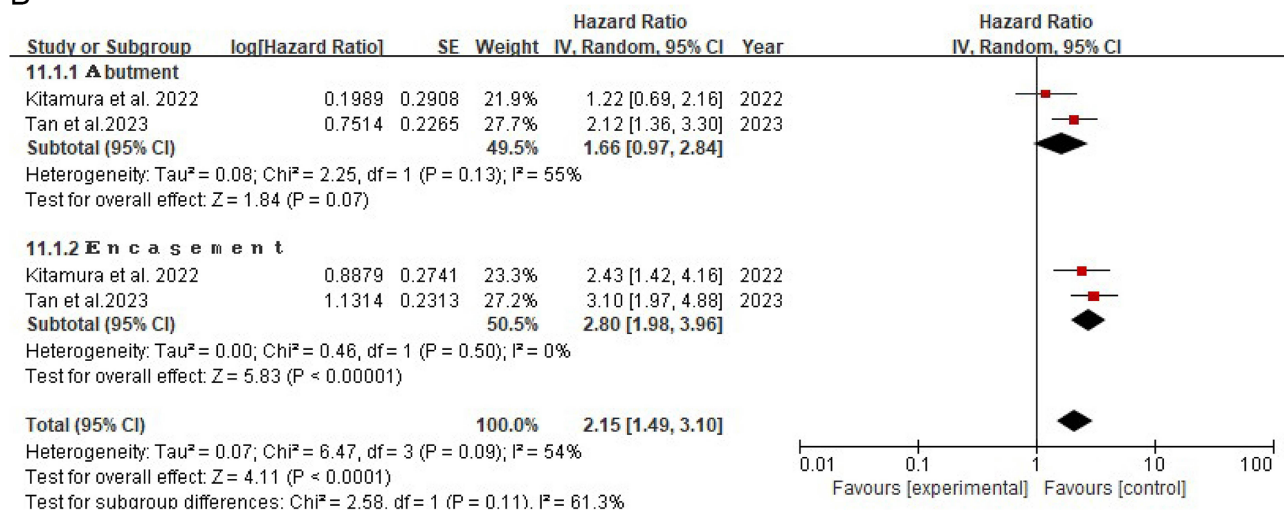


Figure 5. (A) Forest plot of subgroup analysis for the effect of SA on OS, stratified by degree of vascular invasion (Abutment vs. Encasement). (B) Forest plot of subgroup analysis for the effect of SV on OS, stratified by degree of vascular invasion (Abutment vs. Encasement).

were rigorous. Only high-quality retrospective studies were included in this meta-analysis. All selected studies focused on patients who did not receive NAT and had resectable tumors, as the application of NAT for resectable PDAC remains controversial. This homogeneity in the study population reduced the confounding effects of different treatment modalities and tumor resectability status. As a result, it provides a more focused and accurate analysis of the impact of splenic vessel invasion in patients with PDAC of the body and tail.

However, this study has several limitations. First, the relatively small number of studies ($n = 8$) may have limited the statistical power and generalizability of the findings. Second, most of the studies included were single-center studies, with only Yin's being multicenter, which may limit the generalizability of the results due to potential center-specific biases^[18]. Third, variations in the classification of splenic vessel invasion existed across studies. While the majority adopted comparable radiological categories broadly corresponding to *clear/abutment/*

encasement, some reported HRs by grouping abutment and encasement, whereas others analyzed them separately. Such heterogeneity is common in meta-analyses where diagnostic protocols may differ across centers. Additionally, for studies that did not provide HRs directly, we extracted survival data from Kaplan–Meier curves using established methods. Although this approach may introduce a degree of imprecision, it is widely validated and accepted in evidence synthesis^[37–39]. Moreover, the radiological prediction rate of SA/SV involvement may not be consistently high. Although the reported sensitivity and specificity in our study both exceeded 70%, Kawai *et al* reported a sensitivity of only 47.4% in their research, suggesting potential heterogeneity in prediction accuracy across different studies or methodologies^[20] (Supplemental Digital Content Table S2, available at: <http://links.lww.com/JS9/F858>). Importantly, our subgroup and sensitivity analyses consistently supported the robustness of the pooled findings. Given the small sample size and heterogeneity in vessel classification across the included

studies, the possibility of a more nuanced prognostic role for SV invasion cannot be excluded. Future studies with larger cohorts and standardized assessment criteria are needed to better define the clinical significance of individual vessel involvement and optimize treatment strategies for this subgroup of patients.

Conclusion

This meta-analysis demonstrated that splenic vessel invasion is associated with worse survival in patients with PDAC of the body and tail. These findings challenge the current paradigm by suggesting that splenic vessel involvement, even in radiographically resectable tumors, may warrant reclassification as BR subtypes and may serve as a potential indication for NAT.

Ethical approval

Not applicable. This study is a systematic review and meta-analysis based on previously published data. No new patient data were collected, and no ethical approval was required.

Consent

Informed consent was obtained from all participants involved in the study.

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Author contributions

Z.Y.: Study concept and design, data collection, data analysis, interpretation and drafting of the manuscript. J.Z.: Study concept and design, data analysis, interpretation and revision of the manuscript. C.T.: Data analysis, literature review, critical revision of the manuscript. X.L.: Literature screening, quality assessment and data verification. K.D.: Statistical consultation and methodological guidance. Q.T.: Supervision, study design, critical revision of the manuscript and final approval. X.W.: Supervision, critical revision for important intellectual content and final approval of the version to be published. All authors read and approved the final manuscript.

Conflicts of interest disclosure

No relevant conflicts of interest to declare.

Guarantor

Qingquan Tan and Xing Wang.

Research registration unique identifying number (UIN)

This study was registered with PROSPERO (Registration number: CRD420250655019).

Provenance and peer review

Not commissioned, externally peer-reviewed.

Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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